AKAP11 (research)
A-kinase anchoring protein 11
Schizophrenia + cancer

https://www.ncbi.nlm.nih.gov/gene/11215
https://www.ncbi.nlm.nih.gov/nuccore/NM_016248.4

protein?
https://www.ncbi.nlm.nih.gov/protein/NP_057332.1

notes

- AKAP // cyclic AMP
- activates where PKA is in a cell?
- many different receptors–depends on what receptor that AKAP may control PKA in the cell differently
    - changes function of cell
- antipsychotisc block certain receptors
    -
- allen brain atlas (where expressed?)
    - diencephalon (less)
    - mesencephalon
    - somewhat in between at cerebellum
    - pons
    - bleeds over to mylencephalon
-

Give a VERY brief background on the gene and why you chose to study it

• What is the gene? A-kinase anchoring protein 11

The A-kinase anchor proteins (AKAPs) are a group of structurally diverse proteins, which have the common function of binding to the regulatory subunit of protein kinase A (PKA) and confining the holoenzyme to discrete locations within the cell. This gene encodes a member of the AKAP family. The encoded protein is expressed at high levels throughout spermatogenesis and in mature sperm. It binds the RI and RII subunits of PKA in testis. It may serve a function in cell cycle control of both somatic cells and germ cells in addition to its putative role in spermatogenesis and sperm function. [provided by RefSeq, Jul 2008]

Spermatogenesis–when sperm cells are created
https://www.ncbi.nlm.nih.gov/books/NBK553142/
Somatic cells–cells of multicellular organism besides reproductive
Germ cells–the reproductive cells
Cell cycle control–regulate amount of cells? Control lifespan + movement

What is it known to do?
Codes protein

Is it linked to a disease, and are there knockouts of this gene in mice that have similar phenotypes?
Schizophrenia–
https://pubmed.ncbi.nlm.nih.gov/36914870/
"REMARK GeneRIF: Exome sequencing in bipolar disorder identifies AKAP11 as a risk gene shared with schizophrenia."

knockouts–https://www.modelorg.us/portal/article/index/id/10909/post_type/3.html
https://www.cyagen.com/us/en/catalog-model-bank-us/S-KO-05497
https://www.cyagen.com/us/en/catalog-model-bank-us/S-KO-05497
No physical knockouts yet, still in development

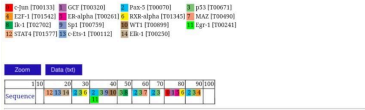
• How is this gene regulated?

```
regulatory      9899..9904
                /regulatory_class="polyA_signal_sequence"
                /gene="AKAP11"
                /gene_synonym="AKAP-11; AKAP220; PPP1R44; PRKA11"
                /note="hexamer: AATAAA"
```

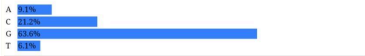
"polyA signal sequence"? signal end
DNA -> mRNA -> protein

• Analyze its promoter
https://epd.expasy.org/epd/search_EPDnew.php?query=AKAP11&db=human
AKAP11_1
gcggcggcggcggcggaagccgaggggtgggcagggcgggggcgggccgcggggctgggagggggagggctgggggtcagcgcgggaggggccagctt

Factors predicted within a dissimilarity margin less or equal than 7 % :



Distribution of the nucleotides over the given chain:



• What transcription factor binding sites are there?

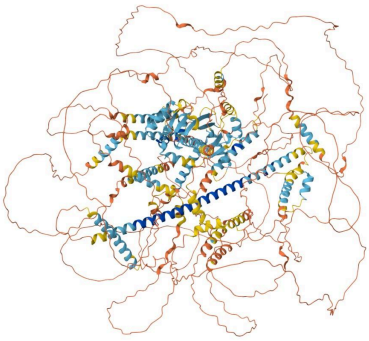
• Have studies shown that this transcription factor actually binds to this promoter or is it just theoretically able to?

• Is there a published protein structure?
No

• If so, include it and highlight which exons form which part of the protein structure

N/A

• If not, use alphafold to make a predicted structure

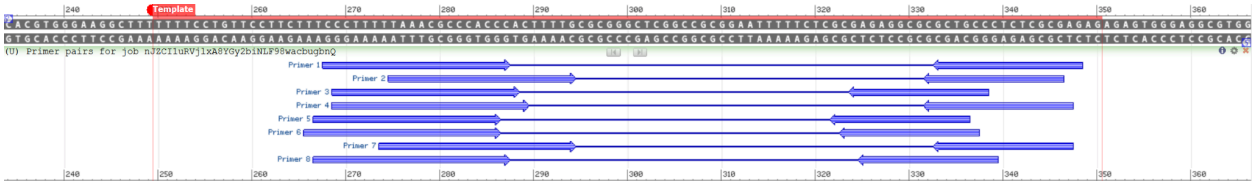


Are there sites in this promoter that are amenable to methylation-specific PCR?
tgaaaccaaatctgcggagcaagccaaggcctggggacttgccagagggaagggttaagccgtttccaaggcaacggaccagga
agcgcgcaagacgtccgccgcaggttcgtaggtcgccctcgaatttaaagctcccaaggctgagtgggtcccggccaatggagagg
cccccttttaagtcatgtgaccagcggggctgccccgcgggctgctccgcgggctgcactggagcacgtgggaaggctttttttcctgttc
cttctttccctttttaaacgcccacccacttttgcgcgggctcggccgcggaatttttctcgcgagaggcgcgctgccctctcgcgagaga
gagtgggaggcgtggagagtaggcggtgttgcgggccggagctgcggcggcggcggcggcggcggaagccgaggggtgggca
gggcgggggcgggccgcggggctgggagggggagggctgggggtcagcgcgggagggggccagctt
https://restrictionmapper.org/
https://restrictionmapper.org/cgi-bin/sitefind3.pl

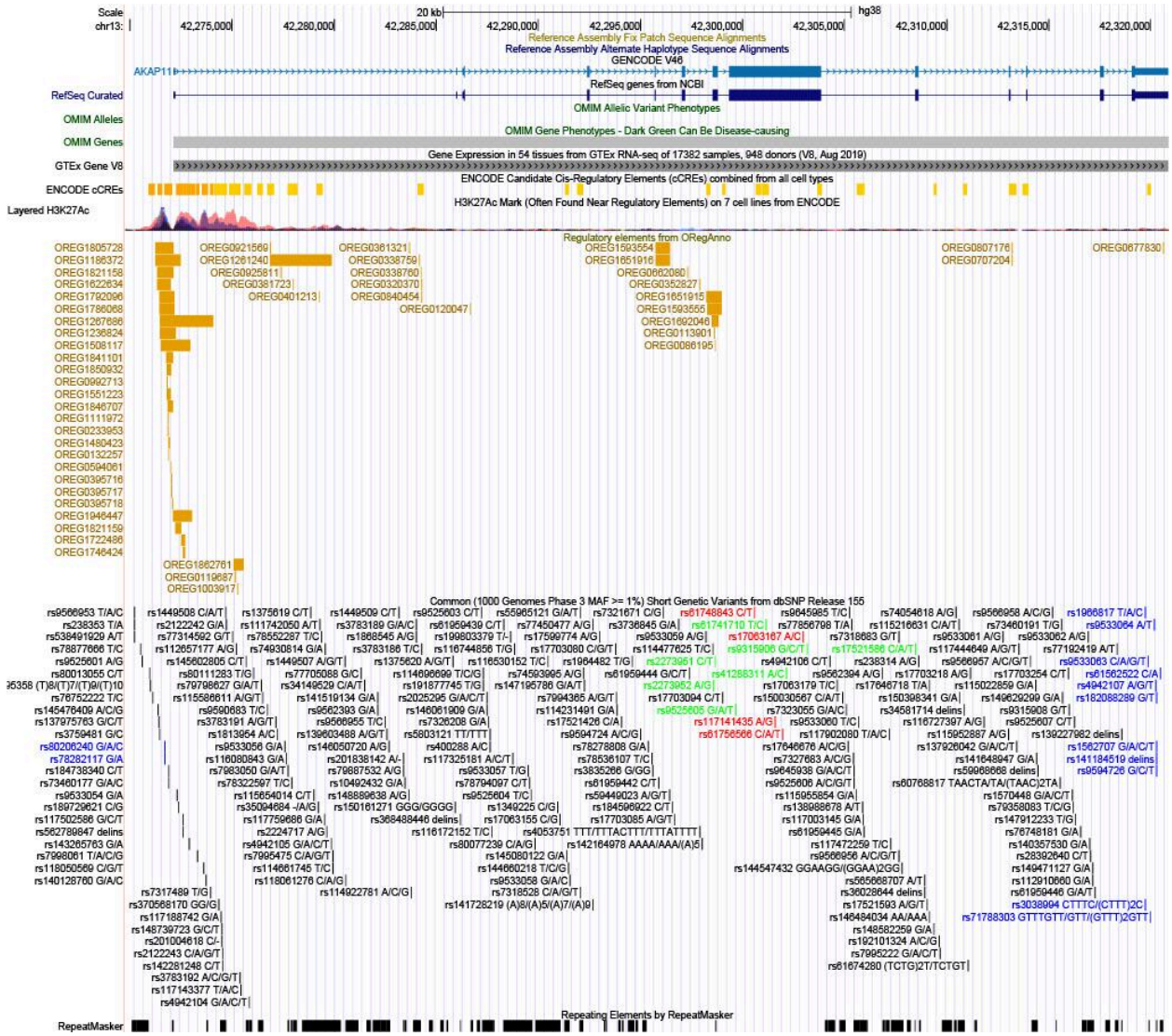
Name: Untitled
Conformation: linear
Enzymes: BstUI
Noncutters:

Name	Sequence	Site Length	Overhang	Frequency	Cut Positions
BstUI	CGCG	4	blunt	10	88, 206, 218, 296, 308, 322, 331, 344, 449, 482

• If so, design primers that would be able to determine this
https://www.ncbi.nlm.nih.gov/tools/primer-blast/primertool.cgi?ctg_time=1722479772&job_key=nJZCI1uRVjlxA8YGy2biNLF98wacbugbnQ



• Find at least one SNP in the promoter region and one SNP in the coding region.
https://genome.ucsc.edu/cgi-bin/hgTracks?db=hg38&lastVirtModeType=default&lastVirtModeExtraState=&virtModeType=default&virtMode=0&nonVirtPosition=&position=chr13%3A42269754%2D42320863&hgsid=2321212104_xLa0FhlxvTKtRaTCyUbX0AoFr0Yx



promoter region doesn't seem to have (red)

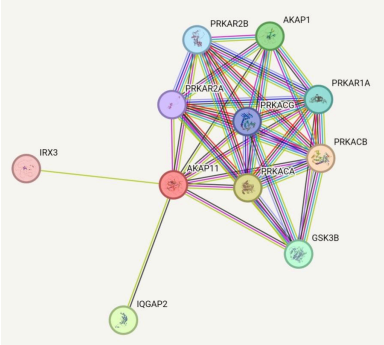
coding region: rs61748843 C/T (Protein-altering variants and splice site variants are red)

MANY SNPs (seemingly), impact multiple places within + mostly spread throughout gene

• Does the promoter SNP impact the predicted transcription factor binding sites?

• For the coding region SNP, find the region in the protein (and the domain) that the SNP impacts

• Use STRING to identify other proteins that your protein interacts with (either directly or indirectly)
https://string-db.org/cgi/network?taskId=bNSGCkbOy2zl&sessionId=bbq44gFyXkSN



• Explain the basis of this interaction
mostly co-expression + textmining, three connections are w/ experimental determination (what is textmining? analysis of texts and stuff)
between the ones relate to AKAP11 they also have interactions between them from curated databases, meaning AKAP11 left out of the loop but the others also interacting stronger (AKAP11 similar processes to more known genes?)
also between the stronger bonds: gene fusions, gene co-occurrence, protein homology
lots of “PRKAR…” and “PRKAC…”

3 interactions
PRKAR2B: **cAMP**-dependent protein kinase type II-beta regulatory subunit
PRKACG:
PRKAR1A:
PRKACB:
PRKACA:
GSK3B:

2 interactions
IQGAP2:
PRKAR2A:
AKAP1: same famiy

1 interaction
IRX3:

• Is there any disease state that you think this gene could be important for?
mental issue of schizophrenia…also seems to be related to bipolar
as well as possibly…cancer? (pubmed articles needed)
issue is likely multiple risk factors besides AKAP11 to cause such neurological diseases

- https://pubmed.ncbi.nlm.nih.gov/37171958/

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articles

- https://pubmed.ncbi.nlm.nih.gov/35410376/
    - AKAP-11 interacts with GSK3B, the hypothesized target of lithium, a primary treatment for BD
- https://pubmed.ncbi.nlm.nih.gov/36914641/
    - eeg mutant mice, akap11 deficiency/mutant parallel schizo
- https://pubmed.ncbi.nlm.nih.gov/37171958/
    - upregulation of proteins associated with autophagy and certain vesicle transport pathways
        - autophagy: A process by which a cell breaks down and destroys old, damaged, or abnormal proteins and other substances in its cytoplasm (the fluid inside a cell). The breakdown products are then recycled for important cell functions, especially during periods of stress or starvation
    - downregulation of proteins related to synaptic, mitochondrial, and ribosomal function in the synapses of individuals with SCZ or BP
    - AKAP11 deficiency again

• Could this gene (or something it interacts with) be a drug target?

• If so, describe a theoretical approach to treat whatever problem you think (or know) that loss-of-function mutations in this gene would have

• Essentially, apply as many of the bioinformatics tools we've covered in camp as possible to learn as much as you can about
the gene and make an organized, visually appealing presentation that shows what you've found

script

- A-kinase anchoring protein 11
- protein-coding (codes AKAP11)
- AKAP family
- is related to the protein kinase A and therefore connected to cyclic AMP
- AKAP may control PKA in the cell differently depending on which receptors activated, can change function of the cell that it is in
- essentially directs where PKA is expressed to change cell function
- schizo risk gene
- allen brain atlas showed more akap11 expression in general areas of mesencephalon and the pons, though it also showed up in the diencephalon, cerebellum, and somewhat in the myelencephalon
    - varied between the subjects that were probed
    - mesencephalon and pons both in midbrain area, mostly motor functions and sensory impulses
- since there was no official structure available, here is what i was able to create with alphafold, the same that was also actually shown on the STRING website
    - thin squiggly lines mean it is extremely unsure of the structuring there, no concrete anything

- knockout mice in development
- not sure if there were any previously, as there had been mutant mice studies, but none currently available

- a few possible transcription factor binding sites, i used 7% for more accuracy compared to the default 15%, so it gave me fewer options

- now on to interactions!

Your Input:

		Neighborhood	Gene Fusion	Cooccurrence	Coexpression	Experiments	Databases	Textmining	[Homology]	Score
AKAP11	*A-kinase anchor protein 11; Binds to type II regulatory subunits of protein kinase A and anchors/targets them; Belongs to the AKAP110 family. (1901 aa)*									

Predicted Functional Partners:

PRKACB	*cAMP-dependent protein kinase catalytic subunit beta; Mediates cAMP-dependent signaling triggered by receptor binding to ...*					●		●		0.948
PRKACA	*cAMP-dependent protein kinase catalytic subunit alpha; Phosphorylates a large number of substrates in the cytoplasm and t...*					●		●		0.939
IQGAP2	*Ras GTPase-activating-like protein IQGAP2; Binds to activated CDC42 and RAC1 but does not seem to stimulate their GTPase...*							●		0.890
AKAP1	*A-kinase anchor protein 1, mitochondrial; Binds to type I and II regulatory subunits of protein kinase A and anchors them to th...*							●		0.857
GSK3B	*Glycogen synthase kinase-3 beta; Constitutively active protein kinase that acts as a negative regulator in the hormonal contro...*					●		●		0.850
PRKAR1A	*cAMP-dependent protein kinase type I-alpha regulatory subunit; Regulatory subunit of the cAMP-dependent protein kinases i...*					●		●		0.817
PRKAR2B	*cAMP-dependent protein kinase type II-beta regulatory subunit; Regulatory subunit of the cAMP-dependent protein kinases in...*					●		●		0.809
PRKACG	*cAMP-dependent protein kinase catalytic subunit gamma; Phosphorylates a large number of substrates in the cytoplasm and...*					●		●		0.775
PRKAR2A	*cAMP-dependent protein kinase type II-alpha regulatory subunit; Regulatory subunit of the cAMP-dependent protein kinases i...*					●		●		0.738
IRX3	*Iroquois-class homeodomain protein IRX-3; Transcription factor involved in SHH-dependent neural patterning. Together with ...*							●		0.731

- STRING interactionns were mostly thru co-expression of genes as well as textmining (using studies to find connections between them)t chromosoe is akap11 on ncbi
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- AKAP11 had only three types of interactions: the aforementioned co-expression and textmining, as well as the addition of experimental determination
- the genes with more interwoven interactions such as PRKAR2B or GSK3B had varying interactions through gene fusions, co-occurence, and protein homology
- strangely enough, the cluster of genes that AKAP11 had the most interactions with (aka having all three mentioned interactions) were the ones tied the most together amongst themselves
- many cAMP related or dependent proteins

- methylation
- used restrictionmapper to get the cut sites
- most of the cuts were clustered around 200-300, more around 300, and therefore i chose to make primers around that area (around 250-350)
- and here are the primers

- i used the ucsc genome browser to get the snps in the gene
- for example, a viable promoter snp might be rs238353, which changes the T to an A, has no reported clinical significance
- a snp in the coding sequence might be rs61748843, which is a C to T snp, which is a benign missense variant
- most utr snps found at the terminator (near end of gene)
- other functional variant snps found around exon 8 of the gene

- risk gene in both schizophrenia and bipolar disorder
- seem to be heavily related
- AKAP11 deficiency as risk (besides protein alteration in both schizophrenia and bipolar)

schizophrenia:

- mutant mice study: AKAP11 deficiency/mutant EEG scans parallel human schizophrenia (GRIN2A also)

bipolar:

- AKAP11 interacts with GSK3B (hypothesized to be targeted by lithium)
- lithium as existing BD treatment–AKAP11 or GSK3B possible drug targets?
- still other possible factors…other genes…many genes tested against both schizo and bd
    - grin2a as a possible yarget, gsk3b being a target for bd eith lithium, since akap11 interacts with both, it cpuld also be affected
- neurological illnesses very complicated
- schizo and bd seem very interconnected

- sources (aka my extremely long bibliography), but i did not format it formally
- epd to get promoter sequence
- articles and ncbi on snps

- the end! thank you for watching
- and…happy end of rosetta